

Moments

The r -th **moment** of a distribution is a measure used to describe the shape of the distribution. It can be measured about an arbitrary point A (raw moment, μ'_r) or about the mean $\bar{X}$ (central moment, μ_r).

or

Example:

- Find the first four moments for the following individual series: 3, 6, 8, 10, 18.

Solution:

Data ($N = 5$): 3, 6, 8, 10, 18.

$$\text{Mean } (\bar{X}) = \frac{3 + 6 + 8 + 10 + 18}{5} = \frac{45}{5} = 9.0$$

Moment	Value
Moments about Mean (μ_r)	$\mu_1 = 0.0$
μ_2 (Variance)	25.6
μ_3	97.2
μ_4	1588.0
Moments about Origin (μ'_r)	μ'_1 (Mean) 9.0
μ'_2	106.6
μ'_3	1517.4
μ'_4	24089.8

or

Example:

- Find the first four moments for the following frequency distribution:

Marks	5-15	15-25	25-35	35-45	45-55	55-65
No. of students (F)	10	20	25	20	15	10

Solution:

Midpoints (m): 10, 20, 30, 40, 50, 60. $N = 100$. $\sum Fm = 3400$.

$$\text{Mean } (\bar{X}) = \frac{3400}{100} = \mathbf{34.0}$$

We calculate the first four central moments (μ_r).

Moment	Formula	Value
μ_1	$\frac{\sum F(m-\bar{X})}{N}$	0.0
μ_2 (Variance)	$\frac{\sum F(m-\bar{X})^2}{N}$	214.0
μ_3	$\frac{\sum F(m-\bar{X})^3}{N}$	468.0
μ_4	$\frac{\sum F(m-\bar{X})^4}{N}$	96712.0



Example:

- The first three moments of a distribution about the value "2" of the variable are 1, 16 and -40. Show that the mean is 3, variance is 15 and the third central moment is -86.

Solution:

Given: Arbitrary point $A = 2$. Raw moments: $\mu'_1 = 1, \mu'_2 = 16, \mu'_3 = -40$.

1. Mean ($\bar{X}$):

$$\bar{X} = A + \mu'_1 = 2 + 1 = \mathbf{3}$$

(Mean is 3. *Shown*)

2. Variance (μ_2):

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 16 - (1)^2 = 16 - 1 = \mathbf{15}$$

(Variance is 15. *Shown*)

3. Third Central Moment (μ_3):

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$$

$$\mu_3 = -40 - 3(16)(1) + 2(1)^3$$

$$\mu_3 = -40 - 48 + 2 = \mathbf{-86}$$

(Third central moment is -86. *Shown*)

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Example:

- The first moments of a distribution about the value "35" are -1.8, 240, -1020 and 144000. Find the values of $\mu_1, \mu_2, \mu_3, \mu_4$.

Solution:

Given: $\mu'_1 = -1.8, \mu'_2 = 240, \mu'_3 = -1020, \mu'_4 = 144000$.

- μ_1 (First Central Moment): By definition, the first moment about the mean is always zero.

$$\mu_1 = \mathbf{0}$$

- μ_2 (Variance):

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 240 - (-1.8)^2 = 240 - 3.24 = \mathbf{236.76}$$

- μ_3 (Third Central Moment):

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$$

$$\mu_3 = -1020 - 3(240)(-1.8) + 2(-1.8)^3$$

$$\mu_3 = -1020 + 1296 - 11.664 = \mathbf{264.336}$$

- μ_4 (Fourth Central Moment):

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

$$\mu_4 = 144000 - 4(-1020)(-1.8) + 6(240)(-1.8)^2 - 3(-1.8)^4$$

$$\mu_4 = 144000 - 7344 + 4665.6 - 34.992 = \mathbf{141290.1072}$$

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Example:

- Calculate the variance and third central moment from the following data:

x	0	1	2	3	4	5	6	7	8
f	1	9	26	59	72	52	29	7	1

Solution:

$$N = \sum F = 256. \quad \sum FX = 1017.$$

$$\text{Mean } (\bar{X}) = \frac{1017}{256} \approx \mathbf{3.9727}$$

1. Variance (μ_2):

$$\text{Variance } (\mu_2) = \frac{\sum F(X - \bar{X})^2}{N} \approx \mathbf{1.9797}$$

2. Third Central Moment (μ_3):

$$\mu_3 = \frac{\sum F(X - \bar{X})^3}{N} \approx \mathbf{0.0179}$$

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Example:

- The first four moments of a distribution about the value '4' of the variable are -1.5, 17, -30 and 108. Find the moments about mean, about origin.

Solution:

Given: $A = 4$, $\mu'_1 = -1.5$, $\mu'_2 = 17$, $\mu'_3 = -30$, $\mu'_4 = 108$.

1. Moments about Mean (μ_r):

- $\mu_1 = 0$
- $\mu_2 = 17 - (-1.5)^2 = 17 - 2.25 = \mathbf{14.75}$
- $\mu_3 = -30 - 3(17)(-1.5) + 2(-1.5)^3 = -30 + 76.5 - 6.75 = \mathbf{39.75}$
- $\mu_4 = 108 - 4(-30)(-1.5) + 6(17)(-1.5)^2 - 3(-1.5)^4$

$$\mu_4 = 108 - 180 + 229.5 - 15.1875 = \mathbf{142.3125}$$

2. Moments about Origin (μ''_r , where $A = 0$):

- μ''_1 (Mean) $= A + \mu'_1 = 4 + (-1.5) = \mathbf{2.5}$
- $\mu''_2 = \mu'_2 + 2\mu'_1 A + A^2 = 17 + 2(-1.5)(4) + 4^2 = 17 - 12 + 16 = \mathbf{21.0}$
- $\mu''_3 = \mu'_3 + 3\mu'_2 A + 3\mu'_1 A^2 + A^3 = -30 + 3(17)(4) + 3(-1.5)(4^2) + 4^3$

$$\mu''_3 = -30 + 204 - 72 + 64 = \mathbf{166.0}$$

- $\mu''_4 = \mu'_4 + 4\mu'_3 A + 6\mu'_2 A^2 + 4\mu'_1 A^3 + A^4$

$$\mu''_4 = 108 + 4(-30)(4) + 6(17)(4^2) + 4(-1.5)(4^3) + 4^4$$

$$\mu''_4 = 108 - 480 + 1632 - 384 + 256 = \mathbf{1132.0}$$

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Example:

- Compute first four moments of the data 3, 5, 7, 9 about mean. Also, compute the first four moments about the point 4.

Solution:

Data ($N = 4$): 3, 5, 7, 9. $\sum X = 24$.

$$\text{Mean } (\bar{X}) = \frac{24}{4} = \mathbf{6.0}$$

1. Moments about Mean (μ_r):

X	$X - \bar{X}$	$(X - \bar{X})^2$	$(X - \bar{X})^3$	$(X - \bar{X})^4$
3	-3	9	-27	81
5	-1	1	-1	1
7	1	1	1	1
9	3	9	27	81
Sum	0	20	0	164

- $\mu_1 = 0.0$
- $\mu_2 = 20/4 = 5.0$
- $\mu_3 = 0/4 = 0.0$
- $\mu_4 = 164/4 = 41.0$

2. Moments about $A = 4$ (μ'_r):

X	$X - 4$	$(X - 4)^2$	$(X - 4)^3$	$(X - 4)^4$
3	-1	1	-1	1
5	1	1	1	1
7	3	9	27	81
9	5	25	125	625
Sum	8	36	152	708

- $\mu'_1 = 8/4 = 2.0$
- $\mu'_2 = 36/4 = 9.0$
- $\mu'_3 = 152/4 = 38.0$
- $\mu'_4 = 708/4 = 177.0$

Example:

- The first four moments about mean of a distribution are 1, 3, 7, 9, 10.

Solution:

The moments about the mean (μ_r) are:

- μ_1 (First Central Moment) is always **0**.
- $\mu_2 = 1$ (Variance)
- $\mu_3 = 3$
- $\mu_4 = 7$

(The remaining values, 9 and 10, are extraneous, as only four central moments exist up to μ_4)

**Example:**

- The first four moments of a distribution about 2 are 1, 2.5, 5.5 and 16 respectively. Calculate the four moments about mean and about the origin.

Solution:

Given: $A = 2$. $\mu'_1 = 1$, $\mu'_2 = 2.5$, $\mu'_3 = 5.5$, $\mu'_4 = 16$.

1. Moments about Mean (μ_r):

- $\mu_1 = 0$
- $\mu_2 = \mu'_2 - (\mu'_1)^2 = 2.5 - (1)^2 = 1.5$
- $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 = 5.5 - 3(2.5)(1) + 2(1)^3 = 5.5 - 7.5 + 2 = 0.0$
- $\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$

$$\mu_4 = 16 - 4(5.5)(1) + 6(2.5)(1)^2 - 3(1)^4$$

$$\mu_4 = 16 - 22 + 15 - 3 = 6.0$$

2. Moments about Origin (μ''_r):

- μ''_1 (Mean) $= A + \mu'_1 = 2 + 1 = 3$
- $\mu''_2 = \mu'_2 + 2\mu'_1A + A^2 = 2.5 + 2(1)(2) + 2^2 = 2.5 + 4 + 4 = 10.5$
- $\mu''_3 = \mu'_3 + 3\mu'_2A + 3\mu'_1A^2 + A^3 = 5.5 + 3(2.5)(2) + 3(1)(2^2) + 2^3$

$$\mu_3'' = 5.5 + 15 + 12 + 8 = 40.5$$

- $$\mu_4'' = \mu_4' + 4\mu_3'A + 6\mu_2'A^2 + 4\mu_1'A^3 + A^4 = 16 + 4(5.5)(2) + 6(2.5)(2^2) + 4(1)(2^3) + 2^4$$

$$\mu_4'' = 16 + 44 + 60 + 32 + 16 = 168.0\$$$

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Example:

- The First four moments of a distribution about are Find the first four moments about mean. Discuss the Skewness and Kurtosis and also comment upon the nature of the distribution.

Solution (Using results from P6/P11 for Skewness and Kurtosis demonstration):

We use the central moments (μ_r) derived from the distribution where $\mu_2 = 14.75$, $\mu_3 = 39.75$, and $\mu_4 = 142.3125$.

1. Skewness (β_1, γ_1):

Skewness measures the degree of asymmetry of a distribution.

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{39.75^2}{14.75^3} \approx 0.492$$

$$\gamma_1 = \sqrt{\beta_1} \approx 0.702$$

Since $\gamma_1 > 0$, the distribution has **Positive Skewness** (or is skewed to the right).

2. Kurtosis (β_2, γ_2):

Kurtosis measures the "peakedness" or "tailedness" of a distribution relative to a normal distribution ($\beta_2 = 3$).

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{142.3125}{14.75^2} \approx 0.654$$

$$\gamma_2 = \beta_2 - 3 = 0.654 - 3 = -2.346$$

3. Comment on Nature of Distribution:

- Skewness:** The distribution is **Positively Skewed** ($\gamma_1 > 0$). The longer tail is towards the positive (right) side.

- **Kurtosis:** Since $\beta_2 < 3$ (or $\gamma_2 < 0$), the distribution is **Platykurtic**. It is flatter than the normal distribution, meaning it has lighter tails and a less pronounced peak.